

Net Force

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Forces in the same direction add, forces in opposite directions subtract, and net force is the overall force with a magnitude and direction.
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Round 1 - Retrieve It

1. How do you combine forces that act in the same direction?

2. How do you combine forces that act in opposite directions?

3. What unit is commonly used for force?

4. What does a net force of 0 N mean about the forces?

Round 2 - Sketch It

Draw a box with a 10 N arrow to the right and a 4 N arrow to the left. Add a net-force arrow and label it.

Net Force - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Two students push a cart right with 5 N and 7 N. Net force?

2. A rope is pulled 18 N east and 11 N west. Net force?

3. A box has 9 N left and 9 N right. Net force?

Round 4 - Reason from Evidence

A cart has 12 N east, 7 N west, and another 5 N west. Find the net force and predict whether the forces are balanced.

Net Force - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Net Force in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect net force to acceleration. What should happen when the net force is not zero?

E**Extend**

Create your own three-force example with a net force of 4 N left. Show the arithmetic.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Net Force

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Add their magnitudes.
2. Subtract the smaller magnitude from the larger and use the direction of the larger force.
3. Newton, N.
4. They are balanced.

Sketch It - look for

Net force is 6 N to the right.

Apply It

1. 12 N right.
2. 7 N east.
3. 0 N; balanced forces.

Reason from Evidence - look for

12 N east versus 12 N west gives 0 N; forces are balanced.

ACE - Extend look-for

Many answers possible. Check that signed/directional forces combine to 4 N left.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.